

# 30-Minute Student Task

## Build Your Own Rule-Based Python Chatbot

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|----------------------|------------------|
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| Programming Language | Python           |

### Assessment :

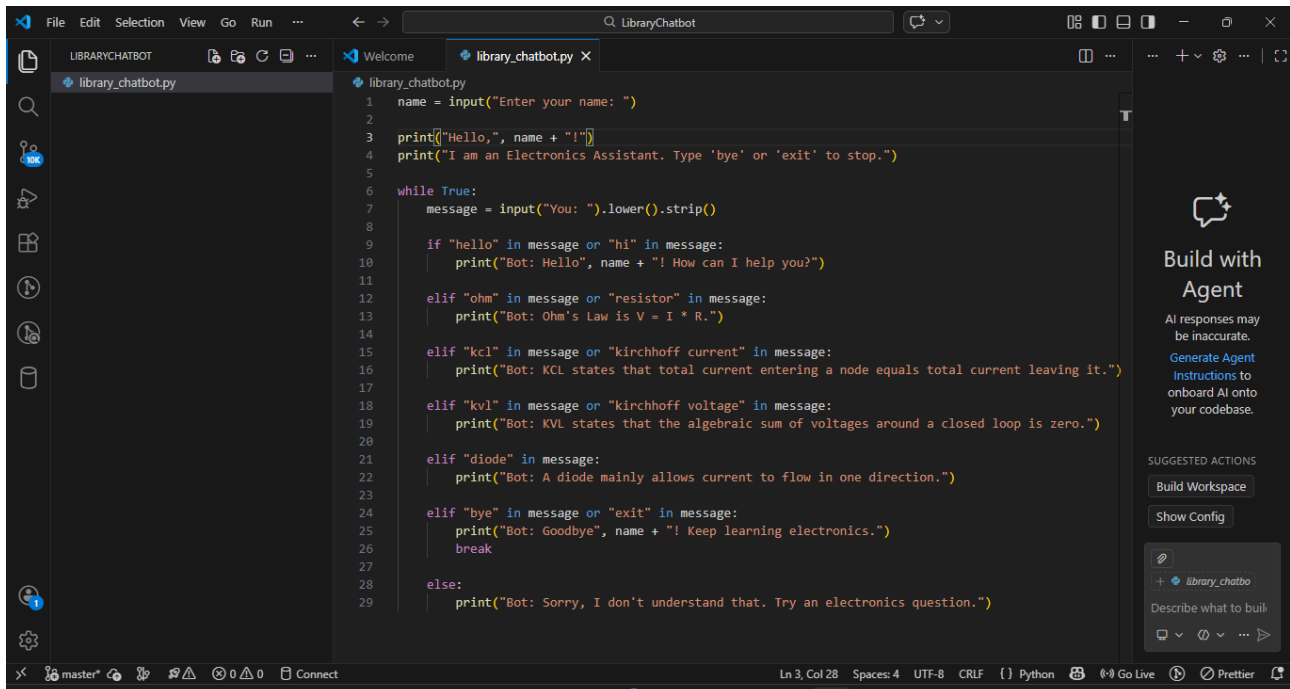
Build a simple rule-based Library Assistant chatbot in Python. The chatbot should interact continuously with the user, recognize greetings and different library-related questions, provide suitable responses, and allow the user to exit the conversation.

### Demonstration Evidence

The following pages contain screenshots of the Python implementation and the terminal execution:

# Python Chatbot Implementation

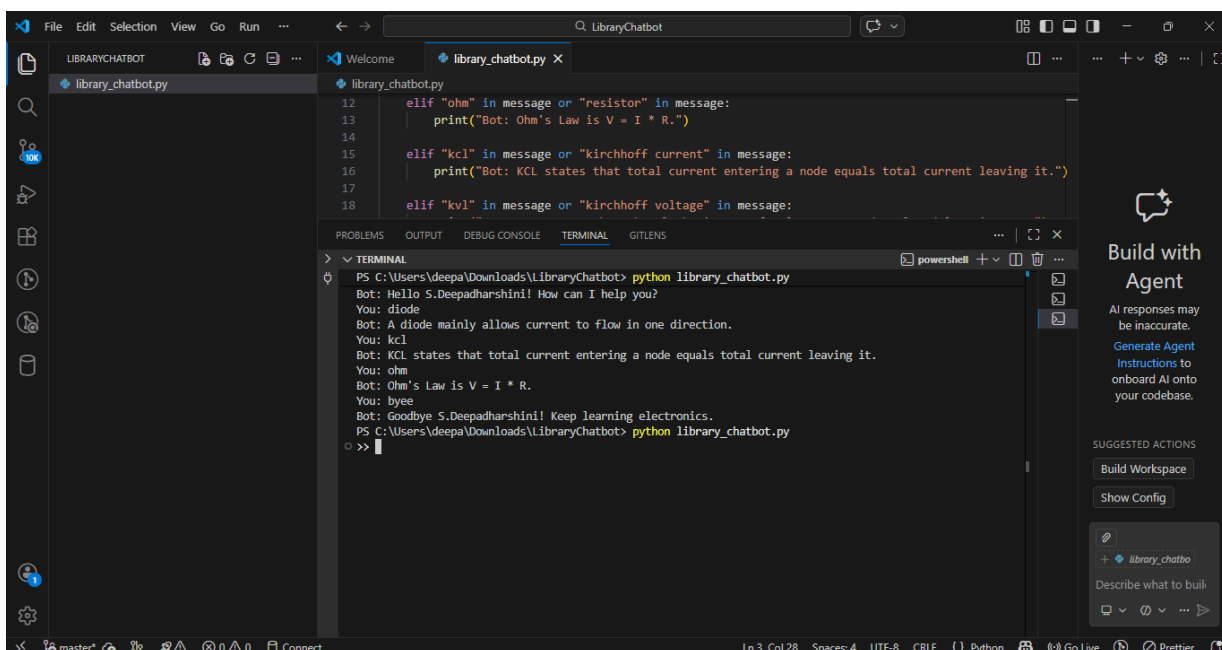
## Screenshot of the Python program implementation



```
1 name = input("Enter your name: ")
2
3 print("Hello,", name + "!")
4 print("I am an Electronics Assistant. Type 'bye' or 'exit' to stop.")
5
6 while True:
7     message = input("You: ").lower().strip()
8
9     if "hello" in message or "hi" in message:
10         print("Bot: Hello", name + "! How can I help you?")
11
12     elif "ohm" in message or "resistor" in message:
13         print("Bot: Ohm's Law is  $V = I * R$ .")
14
15     elif "kcl" in message or "kirchhoff current" in message:
16         print("Bot: KCL states that total current entering a node equals total current leaving it.")
17
18     elif "kvl" in message or "kirchhoff voltage" in message:
19         print("Bot: KVL states that the algebraic sum of voltages around a closed loop is zero.")
20
21     elif "diode" in message:
22         print("Bot: A diode mainly allows current to flow in one direction.")
23
24     elif "bye" in message or "exit" in message:
25         print("Bot: Goodbye", name + "! Keep learning electronics.")
26         break
27
28     else:
29         print("Bot: Sorry, I don't understand that. Try an electronics question.")
```

## Chatbot Execution and Test Output

Terminal output showing greeting, question handling, an unknown message, and the exit command.



```
PS C:\Users\deepa\Downloads\LibraryChatbot> python library_chatbot.py
Bot: Hello S.Deepadharshini! How can I help you?
You: diode
Bot: A diode mainly allows current to flow in one direction.
You: kcl
Bot: KCL states that total current entering a node equals total current leaving it.
You: ohm
Bot: Ohm's Law is  $V = I * R$ .
You: bye
Bot: Goodbye S.Deepadharshini! Keep learning electronics.
PS C:\Users\deepa\Downloads\LibraryChatbot> python library_chatbot.py
o >> |
```